Mining Analytics in the Age of AI

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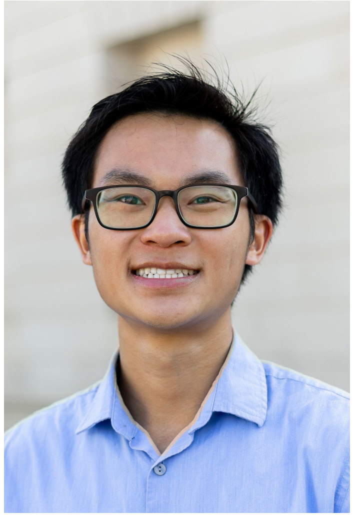
Team Introduction



Donald Thai

Project Lead &
Business Analyst

Mickey Xu

Data Analytics

Sharan Kunusoth

Data Engineer

Austin Frings

Data Architect

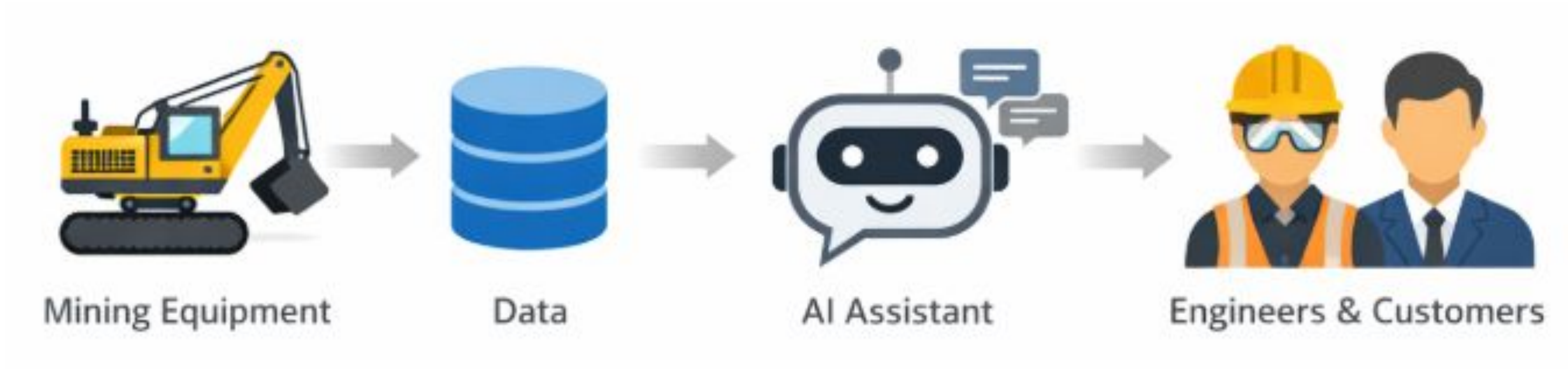
Agenda

1. Introduction to the key business problem
2. Data overview
3. Progress update
4. Q&A

KOMATSU

We Build Machines to Answer the Needs of Society

The focus of our project is to improve the efficiency of gathering data insights

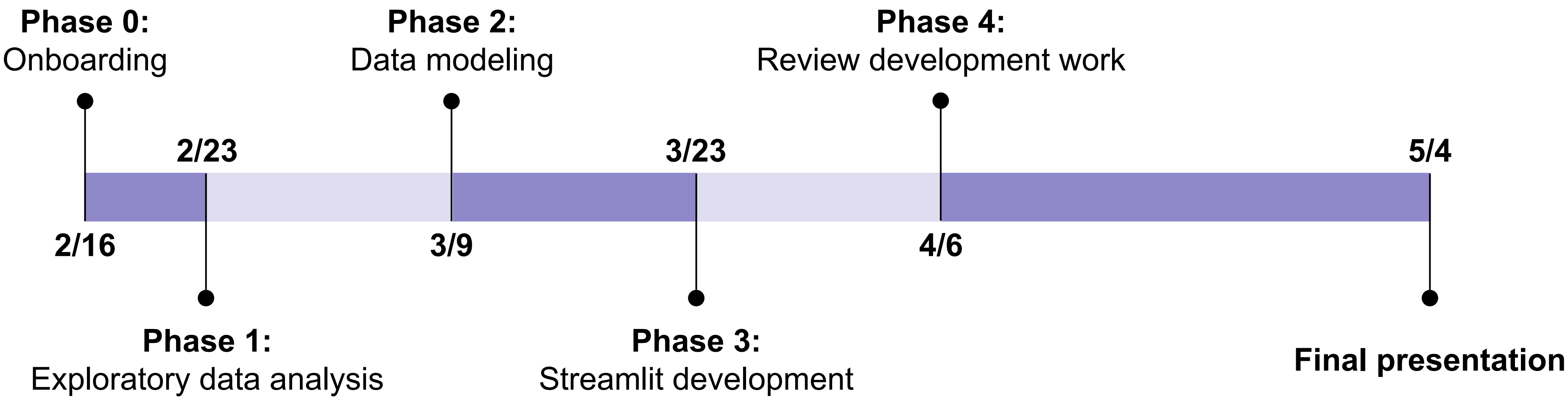


How can we provide instant, reliable, data insights through an AI-powered interface?

The project is split into phases with distinct tasks and deliverables

Deliverables

Execution plan



Coding files for analysis, data modeling, and Streamlit

Our final deliverable is to create a chatbot that answers data insight questions with at least 80% accuracy

Semantic data model using natural language processing

Streamlit AI assistant

Ask a question…

Sample questions

What has been the min, max, and average temperature of 41338's RH shipper shaft bearing over the last month?

What is the average productivity across the 4800 lineup, excluding 48002?

Can you compare propel hours between ES48001 and ES48002 for the month of January?

Must answer specific questions with at least >80% accuracy

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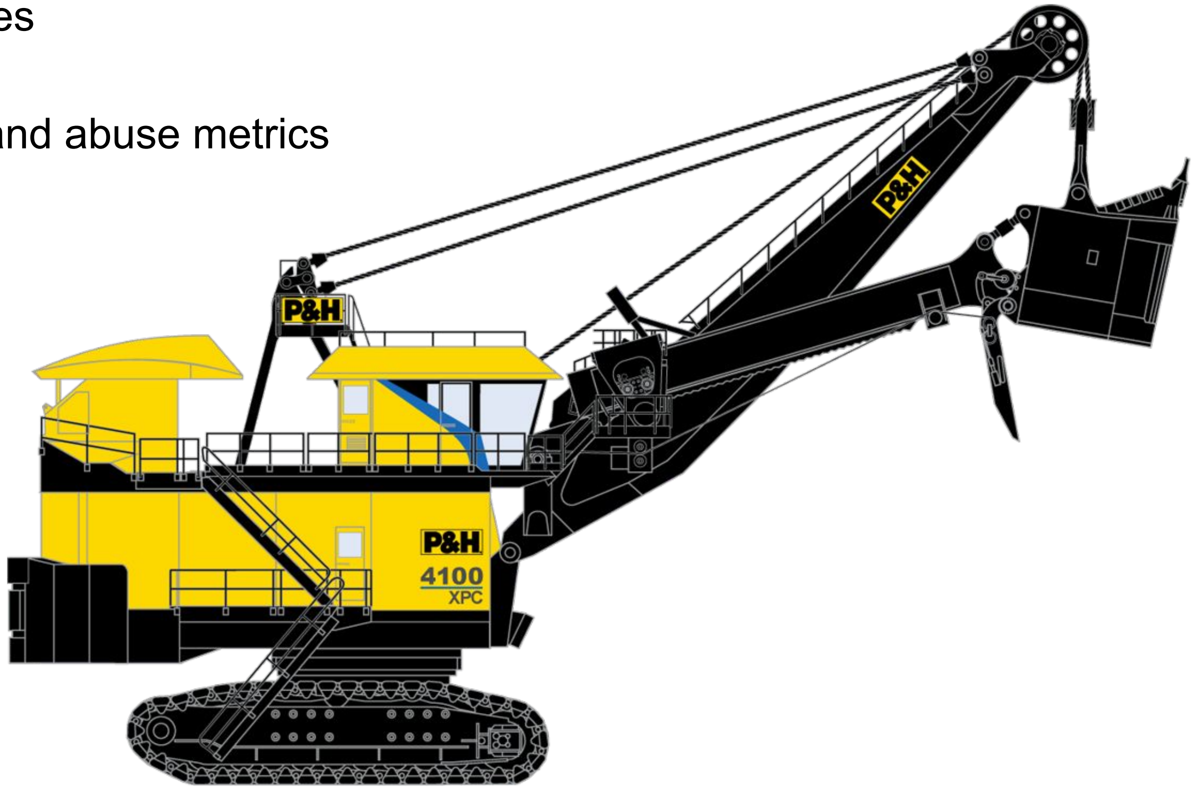
KOMATSU

Data: Equipment Health Events

1. Episodes and Event Codes
2. Shift Report
   a. Shift-level damage and abuse metrics

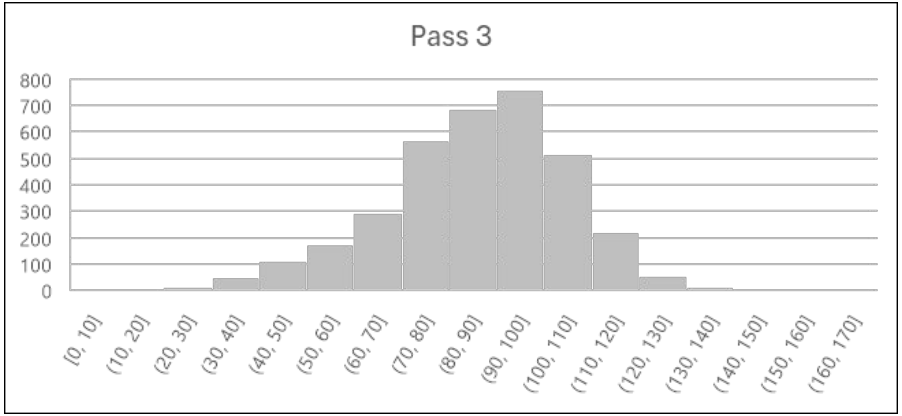


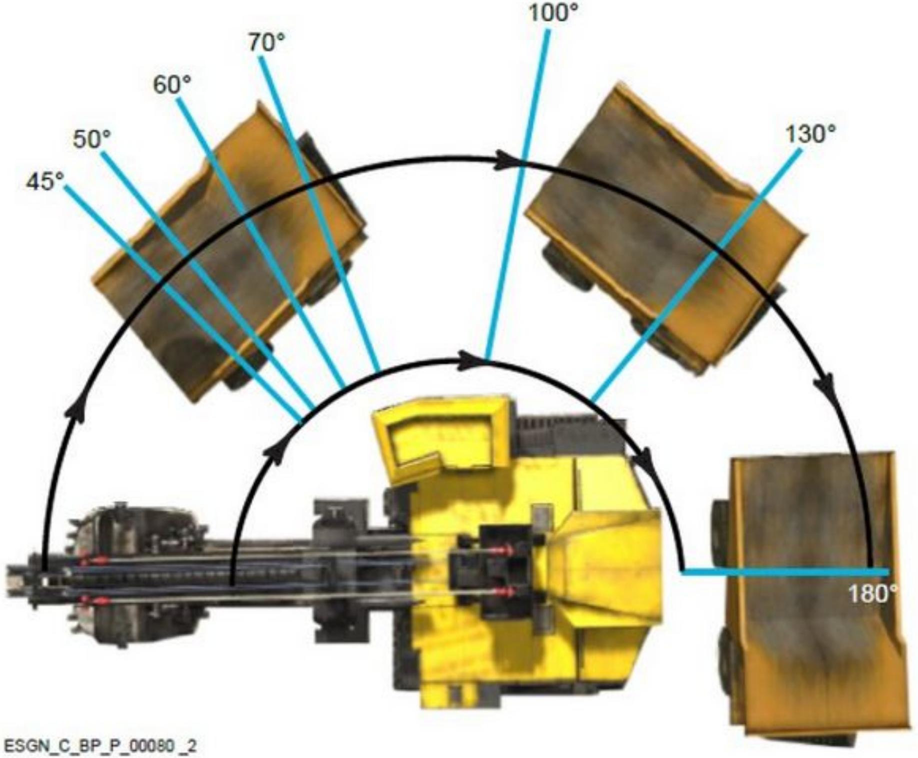
Data: Operations Productivity

1. Shift Report
   a. Dig hours, idle hours, payload amounts, etc
2. 10 Minute Utilization Averages

Data: Dig Cycle Performance

1. Dig Cycles
   a. Timing, payload, and position metrics





We will use Snowflake and Streamlit to perform data analysis, develop a data model, and serve that model to users

Snowflake

Streamlit

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Past, Present and Future- It's All About Community

We are currently exploring the data and creating a prototype for the semantic data model

Phase 1

Explore data in Snowflake

Understand business relations in data

Phase 2

Develop semantic data model

Our next steps include furthering our data analysis and developing a full semantic data model

Phase 1: Data Analysis + Phase 2: Data Modeling	
Date	**Task**
3/16	Continue exploring the data
3/23	Fully develop our semantic data models
3/23	Verify semantic model accuracy

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Appendix

Throughout our project, we will use Snowflake and Streamlit to perform data analysis, develop a data model, and serve that model to users

1. Snowflake
    a. Data warehousing
    b. Secure LLMs
    c. Semantic models for reliable generation

2. Streamlit
    a. Front end application and user interface
    b. Seamless integration with Snowflake

We completed all the onboarding tasks

	Phase 0: Onboarding (2/16 - 2/23)	
	Date	**Task**
✔	2/16	Onboarding meeting
✔	2/18	High level overview
✔	2/19	Meet the application engineering team
✔	2/23	Define data question
✔	2/25	Getting access to the data